

FGV EAESP — SÃO PAULO SCHOOL OF BUSINESS ADMINISTRATION

MSC / PhD IN BUSINESS ADMINISTRATION

Time Series Econometrics for Business Research

Tutorial 4

Case Study: Macro-Finance Linkages: A VAR Analysis of the Brazilian Business Cycle

Application area: *Finance / Macroeconomics — Monetary Policy*

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FRED: BRAGDPNADSMEI (Brazil GDP), BRACPIALLMINMEI (CPI),
INTDSRBRZQ156N (SELIC rate). Also available from BCB (<https://www.bcb.gov.br/en/statistics>). Quarterly, 1996–2024.

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1. Background and Objectives

Understanding how monetary policy shocks transmit to output and inflation is central to macrofinancial analysis. A structural VAR (SVAR) with the SELIC interest rate, output, and inflation provides a standard framework for this analysis.

Learning objectives: (i) Specify and estimate a trivariate VAR; (ii) select lag length and test residual properties; (iii) test Granger causality; (iv) compute and interpret Cholesky-identified IRFs; (v) interpret variance decompositions.

2. Data Preparation

```
library(vars); library(fredr); library(tseries)
# Quarterly: log GDP, CPI inflation rate, SELIC rate
# Build a ts matrix or xts with three columns
y_gap <- log(gdp) - hpfilter(log(gdp))$trend # output gap
infl  <- diff(log(cpi)) * 400                # annualised inflation
selic  <- selic_rate                        # level
VAR_data <- cbind(y_gap, infl, selic)
```

3. Exercises

3.1. Specification and lag selection

- (a) Test each variable for unit roots (ADF + KPSS). Which variables, if any, are $I(1)$?
- (b) Estimate $\text{VAR}(p)$ for $p = 1, 2, 3, 4$. Report AIC, BIC, and HQIC. Select the lag order.
- (c) Apply the multivariate Portmanteau test to residuals at lags 5 and 10. Are residuals white noise?

3.2. Granger causality

Test whether: (a) the SELIC rate Granger-causes the output gap; (b) inflation Granger-causes the SELIC rate (Taylor rule in reverse). Report F -statistics and p -values. Interpret the results in terms of monetary policy transmission.

3.3. Impulse response functions

Identify the VAR using Cholesky ordering: output gap first, inflation second, SELIC third (monetary policy responds last within a quarter).

- (a) Plot the IRF of output gap and inflation to a one-standard-deviation positive shock to the SELIC rate over 12 quarters. Include 90% bootstrap confidence bands.
- (b) Is there a price puzzle (inflation rises after a contractionary shock)? If so, discuss possible causes.
- (c) How does the output gap respond? After how many quarters is the effect largest?

3.4. Variance decomposition

Compute the forecast error variance decomposition (FEVD) of the output gap at horizons 1, 4, 8, and 12 quarters. What fraction of output fluctuations is explained by monetary policy shocks at each horizon?

3.5. Robustness: alternative ordering

Re-identify the SVAR placing the SELIC rate first (monetary policy is predetermined). How do the IRFs change? What does this reveal about the sensitivity of results to identification assumptions?

4. Core References

Sims, C.A. (1980). *Econometrica*, 48, 1–48. Lütkepohl, H. (2005). *New Introduction to Multiple Time Series Analysis*. Springer. Kilian, L. & Lütkepohl, H. (2017). *Structural VAR Analysis*. CUP.
